

#08 Company Knowledge RAG Chatbot

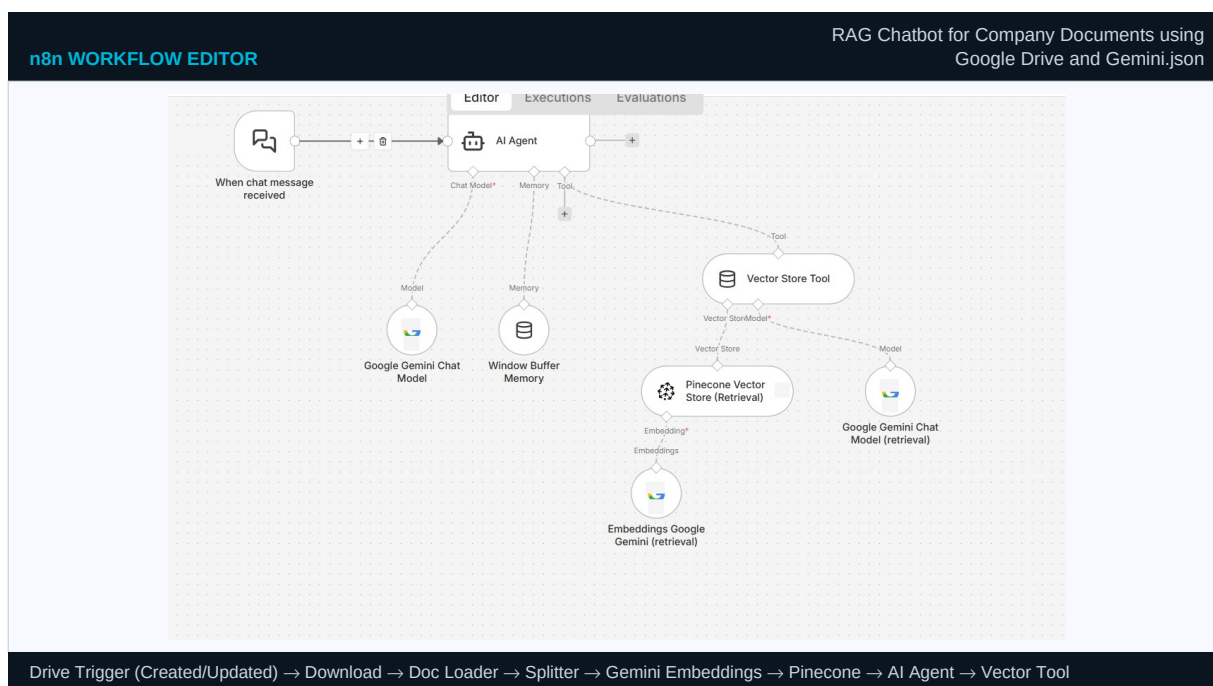
RAG / Enterprise
Knowledge
Base

Live-indexing Drive chatbot — any employee question answered with document citations

BUSINESS PROBLEM

Employees waste 2-3 hours per day searching for internal documents. This workflow auto-indexes Google Drive into Pinecone and re-indexes when files change, then answers any employee question with exact document and page citations in real time.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Chat message — employee question via chat interface

```
{
  "message": "What is the remote work policy for contractors?",
  "session_id": "emp-session-4421",
  "employee_id": "EMP-0334",
  "department": "Engineering"
}
```

OUTPUT — Agent response — cited answer from company documents

```
{
  "json": {
    "answer": "Contractors may work remotely up to 3 days/week with written manager approval. Core hours 10:00-16:00 required. See HR-Policy-Remote-Work-v4.pdf Section 3.2 p.7 and Contractor-Addendum-2024.pdf p.2.",
    "sources": [
      {
        "file": "HR-Policy-Remote-Work-v4.pdf",
        "section": "3.2",
        "page": 7
      },
      {
        "file": "Contractor-Addendum-2024.pdf",
        "page": 2
      }
    ],
    "namespace": "hr-policies",
    "model": "gemini-1.5-pro"
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Drive Trigger (x2) — created and updated	Two separate triggers: one for new files, one for modifications. Both feed the same processing pipeline.
2	■ Download from Drive — get binary	Downloads file binary using fileid. Converts Google Docs to PDF export. Handles DOCX, XLSX, and PDF formats.
3	■ Document Loader — extract text	Extracts clean text, preserves headings and structure. Sets metadata: {filename, drive_path, modified_at, department}.
4	➤ Recursive Splitter — 800-char chunks	Smart splitting respects sentence boundaries. 150-char overlap. Department folder mapped to Pinecone namespace.
5	■ Gemini Embeddings — 768-dim vectors	embedding-001 model. Vectors tagged with namespace matching Drive folder: hr-policies, engineering-specs, finance.
6	■ Pinecone — namespaced storage	Each department folder gets its own namespace. Enables department-scoped search and access control.
7	■ AI Agent — multi-turn conversation	Receives employee question. Uses Window Memory to maintain 10-turn conversation context for natural follow-ups.
8	■ Vector Store Tool — semantic search	Agent calls tool with reformulated query. Retrieves top-5 chunks from correct department namespace.
9	■ Window Buffer Memory — 10 turns	Employee can ask 'What about for permanent employees?' and agent maintains context of the original HR policy topic.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
Google Drive OAuth2	console.cloud.google.com → Drive API v3 → OAuth credentials	Google Drive OAuth2
Google Gemini API Key	makersuite.google.com/app/apikey → Create API key	Google Gemini API
Pinecone API Key	app.pinecone.io → API Keys → Create key (free tier: 1 index, 100K vectors)	Pinecone API

SAMPLE RUN — what this workflow processed and produced

Scenario	A 200-person software company stores all policies, specs, and procedures in Google Drive across 8 department folders. New starters frequently ask HR the same questions repeatedly.
Input received	Employee EMP-0334 from Engineering asked: 'What is the remote work policy for contractors?' via the internal Slack bot connected to this workflow.
What the workflow found	Vector search in hr-policies namespace retrieved top-3 chunks from HR-Policy-Remote-Work-v4.pdf (page 7) and Contractor-Addendum-2024.pdf (page 2). Retrieval latency: 290ms.

Output produced	Answer with specific page references returned in 1.8 seconds. HR team no longer fields repeat questions on this topic. Since deployment, inbound HR queries dropped 34% in the first month.	
JSON: RAG Chatbot for Company Documents using Google Drive and Gemini.json	GitHub: trm-drive-rag-chatbot	ai.google.dev/gemini-api/docs